

CyberConfucius: An Interactive Confucian Philosophical Counseling System for Mental Health

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Abstract

Confucian philosophy offers beneficial insights into modern mental health issues like anxiety and social adaptation confusion, yet its value is often locked behind archaic texts. We present CyberConfucius, an LLM-based, culturally grounded interactive counseling system that brings Confucian wisdom to life in the form of philosophical counseling to help promote mental health. The system proposes a novel idiom-based retrieval-argument generation strategy to map contemporary user counseling questions with quotes from *The Analects*, and leverages prompts to provide empathetic responses, proactive questioning, and structured summaries to guide deep philosophical reflection. A preliminary user study suggests promising potential to align LLMs with philosophical classics for mental health support.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

Philosophical counseling; Confucian; Large language model; Retrieval-augmented generation

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1 Introduction

The rapid transformation of modern societies has precipitated a global mental health crisis. Mental health disorders such as stress, anxiety, and existential despair now prevail all over the world. Even as material living standards improve, individuals continue to face a deep void in spiritual fulfillment and self-understanding [9, 15, 16]. According to the World Health Organization [14] and corroborated by global burden studies [13], the prevalence of mental health challenges demands innovative and integrative approaches that extend beyond conventional therapeutic models.

In recent years, contemporary philosophical counseling has emerged as a promising paradigm to address mental challenges. Unlike traditional psychological counseling, which often focuses on symptom alleviation, philosophical counseling encourages critical self-reflection and inquiry, prompting individuals to revisit and reassess their core values and life purposes [2, 5]. The reflective dialogue not only offers relief from immediate distress but also provides more fundamental and lasting therapeutic effects. Despite the effectiveness of philosophical counseling for mental well-being, its reach is currently limited [11]. Existing practices are largely Western, with little application in Eastern cultures. [3]. Besides, philosophical counseling still largely depends on paid human-practitioner services, which restricts its scalability and accessibility.

Well-established findings show that culture has long served as an effective means for societies to mitigate mental crises [7], even extreme threats like death anxiety, by offering a sense of order and meaning [6]. Confucian wisdom, deeply rooted in Eastern culture across centuries, has provided a robust spiritual framework with values such as *ren* (benevolence), *yi* (righteousness), *li* (propriety), *zhi* (wisdom), and *xin* (trustworthiness) [4, 8]. Particularly, as recorded in *The Analects*, Confucian philosophy is presented in the form of dialogues. In this context, the rich canon of Confucian philosophy inherently holds significant potential for philosophical counseling to address modern existential concerns, from the perspectives of both the conversational format and the cultural contents.

With remarkable conversational capabilities, large language models (LLMs) offer unprecedented opportunities to automate the service pipeline and facilitate the promotion of philosophical counseling [3]. In this paper, by integrating classical Confucian wisdom

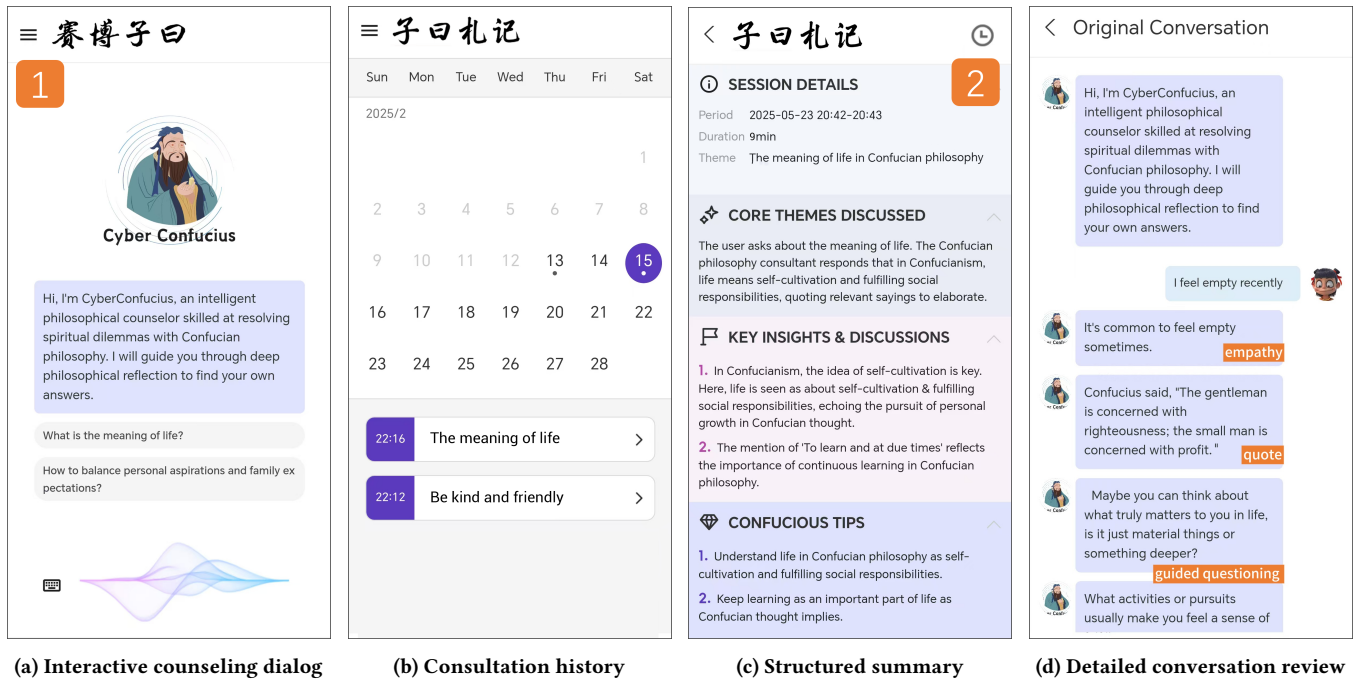


Figure 1: The Main Interfaces of CyberConfucius System.

with contemporary AI techniques, we propose CyberConfucius, an interactive Confucian philosophical counseling system powered by LLMs to improve mental health, which embodies two central innovations. Firstly, CyberConfucius aligns Confucian paradigms from *The Analects* in archaic texts with user consultation topics in the modern Chinese language through a retrieval mechanism based on thematic four-character idioms. Accounting for the linguistic nuances between the usage of the ancient and modern Chinese language, we leverage four-character thematic idioms which maintain ancient language style but are still widely used in modern language to enhance the Confucian comprehension of LLMs. The idiom-based Retrieval-Augmented Generation (RAG) framework is then novelly designed to bridge classical Confucian texts and user daily life queries, enabling culturally grounded semantic matching and content generation. Each response comprehensively understands users' inquiries, cites precise original texts from *The Analects*, and provides daily explanations, combining both ancient and contemporary insights of Confucian philosophy. Such meticulous alignment ensures the system remains authentic in Confucian foundations and meanwhile practical in daily application. Secondly, targeted prompt engineering is designed to facilitate CyberConfucius with typical conversational styles of philosophical counseling, where CyberConfucius is able to express empathetic responses to users and emulate the reflective questioning as human counselors.

Moreover, our system provides an intuitive and interactive experience. It supports multimodal interaction, allowing users to interact with the system not only through text but also via voice input. By prompting LLMs, it also provides structured counseling summarization to further motivate deeper introspection and reflection. Developed as a web-based application, CyberConfucius

can seamlessly adapt to diverse devices, which makes philosophical counseling more accessible. These design elements collectively create an interactive and immersive counseling environment that promotes self-reflection and personal growth through guided Confucian philosophical dialogues.

By merging ancient wisdom with cutting-edge AI, CyberConfucius stands as a transformative system for contemporary mental health support, offering both profound theoretical insights and wide-ranging practical benefits as follows.

- **Innovative integration.** To the best of our knowledge, this work is the first to integrate LLMs with traditional Confucian philosophy to facilitate intelligent philosophical counseling for mental health promotion, which provides a culturally grounded and interactive modality that complements and extends traditional psychological counseling paradigms.

- **Technical innovation.** The novelly designed idiom-based RAG strategy contributes to the alignment of LLMs' comprehension between ancient and modern Chinese language, bridging ancient wisdom and modern mental health concerns. This framework holds promising potential in other similar tasks that strengthen LLMs' capabilities of understanding and utilizing Chinese classics in daily contexts.

- **Theoretical advancements.** The CyberConfucius system offers novel insights into how cultural and philosophical traditions can be mobilized to address existential anxieties, enriching the academic discourse on mental health, philosophical practice, and cultural revitalization.

- **Practical implications.** As our work demonstrates how technology can be harnessed to revive traditional culture and promote

a more holistic approach to well-being, it paves the way for advancements in human–computer interaction and the development of AI-based mental health interventions.

2 System Design

2.1 Idiom-based RAG from *The Analects*

Main idea. CyberConfucius introduces an idiom-based RAG framework that uses four-character idioms as a semantic bridge between modern user queries and the archaic language of *The Analects*. Given a sequential dialogue piece in the consecutive conversation Q , the goal is to retrieve the most relevant quote from *The Analects* to improve the quality of Confucian philosophical counseling. Due to the limitations of general LLMs in understanding archaic Chinese, CyberConfucius innovatively exploits four-character idioms, which are more concise and could be better understood by LLMs, as a bridge to align the archaic texts of *The Analects* with users' modern Chinese expressions through retrieval-augmented generation. Table 1 illustrates the framework of the alignment. Users' daily counseling topics such as emotional regulation, marital challenges, social role identity, and global justice can be mapped to Confucius's philosophical framework of self-cultivation, family management, society and governance, and harmony with the whole world. For the counseling example "how can we build a more sustainable world", CyberConfucius first utilizes an LLM to generate semantically similar idiom such as "天人合一" (the unity of heaven and humanity), and then locates the most semantically similar content in *The Analects* "Confucius fishes but not with a net, and shoots but not at a bird at rest" as the counseling response.

To construct the idiom-based RAG framework, we first established a curated set of four-character idioms and their corresponding mappings to quotes from *The Analects*. f_θ denotes inference LLMs like DeepSeek-R1, responsible to generate the backend idiom-quote pairs, while f_ϕ denotes interaction LLMs like DeepSeek-V3, which are employed for real-time consultation dialog.

Idiom-quote association construction. For each original Analects quote c_i and the corresponding expert interpretation text t_i , an LLM f_θ evaluates their intellectual and philosophical significance in modern contexts via prompts, producing a 5-point Likert score $j_i = f_\theta(c_i, t_i)$, where a bigger score suggests a higher significance in modern contexts. Based on the score j_i and the corresponding interpretation t_i , f_θ then generates $|j_i|$ thematic four-character idioms for the quote pair (c_i, t_i) , denoting as a set of thematic idioms $\mathcal{P}_i = \{p_{i1}, \dots, p_{ij_i}\}$. For each idiom p_{ij} , the model f_θ also outputs a semantic justification r_{ij} describing its relevance to (c_i, t_i) . All generated thematic idioms are merged into a global set $\mathcal{P}^* = \bigcup_i \mathcal{P}_i$. An inverted index is built to map each thematic idiom $p \in \mathcal{P}^*$ to all its associated quote entries $R(p) = \{c_i \mid p \in \mathcal{P}_i\}$. Specifically, to ensure a high-level semantic focus on the most relevant idiom-quote pair, we set the constraint that each idiom corresponds to at most 5 quotes (i.e., $|R(p)| \leq 5$).

Real-time retrieval augmentation. Within an ongoing counseling session with CyberConfucius, the pipeline of real-time retrieval-augmented generation is shown in Fig. 2. When the user responds with a new conversation piece, the multi-turn dialogue Q is passed into f_ϕ . The f_ϕ model is responsible for generating an idiom set

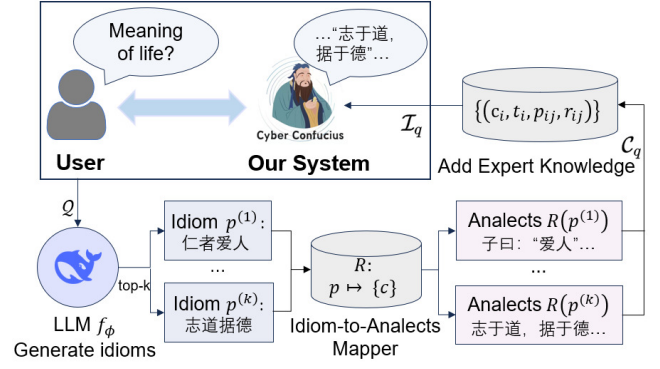


Figure 2: The thematic 4-character idiom based RAG.

$\mathcal{P}_q = \{p^{(1)}, \dots, p^{(n)}\} \subseteq \mathcal{P}^*$, representing the top- k contextually relevant and inspirational idioms given the dialog Q . Using selected idioms $p^{(1)}, \dots, p^{(k)}$, the system retrieves the associated classical entries $C_q = R(p^{(1)}) \cup \dots \cup R(p^{(k)})$. In the implementation of our system, we set $k = 2$. Each entry $c_i \in C_q$ and idiom $p_{ij} \in \mathcal{P}_q$ is associated with metadata $\{(t_i, r_{ij})\}$. These are aggregated into an enriched input set: $\mathcal{I}_q = \{(c_i, t_i, p_{ij}, r_{ij}) \mid c_i \in C_q, p_{ij} \in \mathcal{P}_q\}$. The enriched dataset is then passed to the philosophical counseling module, which integrates the quote from *The Analects*, modern interpretations, and semantic annotations to generate stylistically aligned and interpretively rich responses.

2.2 Philosophical Counseling Format Control

To emulate a human consultant, the system's responses are controlled to include three characteristics of philosophical counseling: *empathetic expressions*, *Confucian philosophy quotes*, and *continuous guiding questions to inspire philosophical reflection* (shown in orange annotations in Fig. 1d). Prompt engineering is leveraged to control such counseling format. System prompts instruct the model to provide guidance and inquiry grounded in Confucius philosophy rather than direct solutions. Specifically, given a user's question, the model is required to refer to the most suitable quotes from *The Analects* and discuss with the user. The reference content is further supported by our RAG strategy. Besides, as philosophical counselors frequently conclude their responses with questions to facilitate further reflection, our model is also instructed to exhibit in a similar way, ending its responses with a question when engaging users in dialogue. The generation temperature is set to 0.5 to encourage diversity and creativity in responses. The model supports memory over 50 dialogue turns, ensuring semantic continuity and depth in extended conversations.

2.3 Consultation Summary Generation

As reviewing and summarizing the consultation positively enhances users' perception of the consultation coherence [10], CyberConfucius generates a structured consultation summary to strengthen users' self-efficacy and engagement in philosophical reflection. The summary is automatically conducted over an integrated counseling session. A new session begins when a user starts a new conversation or after 30 minutes of inactivity.

Table 1: Confucian Framework Applied to Philosophical Counseling

User Counseling Topics	Example Question	Confucian Level	Corresponding Thematic Idiom	Quote from <i>The Analects</i>
Self-discipline, emotional regulation, pursuit of meaning	How can I find the true meaning and purpose of my life?	Individual: Self-Cultivation	修身养性 (Cultivate oneself and nourish one's nature)	Self-introspection should be practiced every day.
Generational gaps, filial duties, marital challenges	How can I handle conflicts among family members?	Family: Family Regulation	和而不同 (Harmony in diversity)	There are differences in harmony gentleman.
Civic responsibility, leadership dilemmas, social role identity	As a citizen, how should I actively fulfill my social responsibilities and obligations?	Nation: Society and Governance	正己守道 (Rectify oneself and uphold the Way)	A wise man devotes his attention to what is essential in the foundation of life.
Environmental ethics, global justice, cultural continuity	How can we build a more sustainable world?	World: Harmony with the Whole World	天人合一 (Unity of heaven and humanity)	Confucius fishes but not with a net; he shoots, but not at a bird at rest.

Based on the detailed contents of a session, CyberConfucius leverages Deepseek-V3 to generate a structured summary by carefully designed prompts. As shown in Fig. 1c, the summary consists of four parts: 1) Session details, recording the theme, timestamps, and duration of the corresponding counseling; 2) Core theme discussed, briefly summarizing the content discussed between the user and the system; 3) Key insight and discussions, which concludes the key points during the consultation; 4) Confucius tips, delivering take-away suggestions for users from the Confucian philosophy. All generated content is designed to be concise and clear. Particularly for key insights and Confucius tips, the model is constrained to generate a maximum of 3 items each. In addition to the summarization, the system also provides users with access to the original conversation, enabling deeper review and reflection.

3 Demonstration

Our system is developed as a web application, lightweight and compatible with various mobile devices, such as smartphones and tablets. Users can access CyberConfucius either as anonymous guests or by registering and logging in. The anonymous guest mode does not save dialogues, while registered users can access their counseling history and summaries.

When the user enters the system, an initial counseling session is automatically created (Fig. 1a). The user can freely express their mental dilemmas and seek philosophical assistance from CyberConfucius via voice, simulating a human-like experience. An alternative text-based interaction is also available by clicking the keyboard icon at the bottom left of the interface. During the conversation, the system acts as a Confucian philosophical consultant, responding in an empathetic tone with quotes from *The Analects*. It also proactively engages users with guided questions to encourage sufficient self-expression and deeper philosophical reflection (shown as the contents in Fig. 1d).

By clicking the main menu (Fig. 1a ①), users can register and log in with individual accounts. After logging in, the personalized counseling history could be archived and the system will provide structured counseling summarizations. Users can review their consultation history by calendar (Fig. 1b). Clicking into a certain consultation record, the user will receive a structured summary of the

consultation, including session details, core themes discussed, key insights, and special Confucius tips as mental health suggestions (Fig. 1c). Users can also access the original conversation (Fig. 1d) by clicking the corresponding button in Fig. 1c ②.

4 User Study

We conducted a preliminary user study to investigate the user experience of CyberConfucius, with a total of 455 participants (286 females, 169 males) recruited. After an initial experience with the system, participants completed a user experience questionnaire, including perceived usefulness, perceived ease of use, information quality, self-efficacy, and intention to use [1, 12]. Each dimension consists of 3 items, resulting in a total of 15 items. The questionnaire adopted a 5-point Likert scale, with a range from *strongly disagree* (= 1) to *strongly agree* (= 5). The Cronbach's α coefficients of all dimensions exceeded 0.7, indicating high internal consistency.

Comparing against the neutral score of 3, One-sample t-tests revealed that users significantly perceived positive and satisfying experience in Perceived Ease of Use ($M = 4.26$, $SD = 0.67$, $p < .001$), Information Quality ($M = 4.27$, $SD = 0.58$, $p < .001$), Perceived Usefulness ($M = 4.32$, $SD = 0.61$, $p < .001$), Self-Efficacy ($M = 4.19$, $SD = 0.62$, $p < .001$), and Intention to Use ($M = 3.81$, $SD = 0.79$, $p < .001$), respectively, validating the effectiveness of our system design.

5 Conclusion

This paper presents CyberConfucius, an easy-to-use Confucian philosophical counseling system designed to provide users with accessible, interactive daily philosophical guidance grounded in classical Chinese wisdom. The preliminary user study indicates the promising potential of LLMs in delivering culturally informed, value-oriented humanistic mental health support. Future research could explore alternative strategies to align LLMs with cultural classics, and comprehensive user experiments to examine performance on mental wellbeing promotion.

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